Comparative Philosophy Volume 17, No. 2 (2026): 4-17
Open Access / ISSN 2151-6014 / www.comparativephilosophy.org
https://doi.org/10.31979/2151-6014(2026).170205

THE MODAL CATUṢKOṬI: FORMALIZING EMPTINESS IN CLASSICAL MODAL LOGIC

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ABSTRACT: *Nāgārjuna's arguments in the* Mūlamadhyamakakārikā *rely extensively on modus tollens, yet this inference is invalid in the paraconsistent frameworks typically used to formalize the* catuṣkoṭi*. We develop a modal extension of Belnap-Dunn semantics to address this, expressed as: ¬□(v(P) = t) ∧ ¬□(v(P) = f) ∧ ¬□(v(P) = b) ∧ ¬□(v(P) = n). This formula applies self-reflexively, resolves longstanding interpretive debates regarding verses within Nāgārjuna's* Mūlamadhyamakakārikā *and* Vigrahavyāvartanī *such as his "no thesis" defense, and bridges interpretive gaps between traditionally designated Eastern and Western logical traditions, providing a Western analytical translation of Indian dialectic into symbolic modal logic that is system-agnostic across K, T, S4, S5, non-normal logics, and neighborhood semantics. The modal operators function classically while the four-valued structure is modalized, preserving modus tollens at the level of modal reasoning while accommodating paraconsistent valuation at the object level. Self-application produces a non-terminating regress that is virtuous rather than vicious: each application is self-contained, demonstrating the unavailability of stable ground rather than constituting a failure to find it.*

Keywords: catuṣkoṭi*, Madhyamaka, modal logic, Nāgārjuna, paraconsistent logic,* svabhāva

1. INTRODUCTION

Central to philosophical Mahayana and Vajrayana Buddhism is the Madhyamaka tradition, traditionally held as founded by the ancient Indian thinker Nāgārjuna. Madhyamaka incorporates a distinctive argumentative and analytical structure known as the *catuṣkoṭi*, or *tetralemma* in order to discuss the concept of '*svabhāva*' – inherent metaphysical essence or existence independent from outside alteration. The *catuṣkoṭi* consists of four sides, or lemmas: [X] has svabhāva; [X] does not have *svabhāva*; [X] both does and does not have *svabhāva*; [X] neither has nor does not have *svabhāva*. *Svabhāva*, as inherent existence or essence independent of causes and conditions, is intrinsically a modal property; to possess *svabhāva* is to possess one's nature necessarily (Westerhoff 2009, 22).

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Graham Priest in "The Logic of the Catuskoti" (2010) has explored how the *catuṣkoṭi* operates within paraconsistent four-valued logic (using true, false, both, and neither) to avoid the consequences of the principle of explosion in classical and intuitionistic logic. However, Nāgārjuna made extensive use of *modus tollens*, which is invalid under Priest's paraconsistent four-valued logics. This paper offers a modal extension of Belnap-Dunn semantics to resolve this problem. By reasoning classically to refute modal necessity-claims, we can employ *modus tollens* without restraint in a system-agnostic fashion, valid across K, T, S4, S5 and extending naturally to non-normal modal logics and neighborhood semantics, where analogous results hold under their respective interpretations of necessity.

We express this as the formula:

$$\neg\Box(v(P) = t) \wedge \neg\Box(v(P) = f) \wedge \neg\Box(v(P) = b) \wedge \neg\Box(v(P) = n)$$

The modal operators function classically; the four-valued structure is what is being modalized. This preserves *modus tollens* at the level of modal reasoning while accommodating paraconsistent valuation at the object level. By negating the necessity claims of all truth-value assignments, the formula achieves self-application, embracing the lack of a halting condition to analysis. This provides an internally consistent explanation for Nāgārjuna's statement within the *Vigrahavyāvartanī* (*The End of Disputes*) in which he seems to assert the absence of a thesis. The formula, as generalized method, applies to every positive assertion - including the assertion of the formula itself - without reaching contradiction or explosion.

2. THE FORMULA

2.1 COMPONENTS AND INTERPRETATION

It is necessary to define the components of the formula for the purposes of disambiguation.

Let P be any proposition.

Let v be a valuation function assigning truth values to propositions, where $v(P) \in \{t, f, b, n\}$:

- t: true (supported by evidence, not refuted)
- f: false (refuted by evidence, not supported)
- b: both (supported and refuted)
- n: neither (neither supported nor refuted)

These are the four values of Belnap-Dunn semantics (Belnap 1977; Dunn 1976), originally developed for reasoning with incomplete and inconsistent information.

Let $\Box$ denote the necessity operator of standard modal logic, where $\Box\varphi$ asserts that φ holds necessarily (in all accessible worlds under Kripke semantics, or under analogous interpretations in other frameworks).

Let $\neg$ denote classical negation; let $\wedge$ denote conjunction.

2.1.1 FORMATION RULES

The well-formed formulas of the language are defined recursively as follows:

1. If P is a propositional variable and $x \in \{t, f, b, n\}$, then $v(P) = x$ is a formula.
2. If φ is a formula and $x \in \{t, f, b, n\}$, then $v(\varphi) = x$ is a formula.
3. If φ is a formula, then $\Box\varphi$ is a formula.
4. If φ is a formula, then $\neg\varphi$ is a formula.
5. If φ and ψ are formulas, then $\varphi \wedge \psi$ is a formula.
6. If φ and ψ are formulas, then $\varphi \rightarrow \psi$ is a formula.

Propositional variables enter the language only through valuation atoms $v(P) = x$. No formula asserts P independently of a valuation context, reflecting the dependence structure central to the Madhyamaka analysis and preventing *svabhāva*-claims at the level of P assertions.

We define $\varphi \vee \psi$ as $\neg(\neg\varphi \wedge \neg\psi)$ and $\Diamond\varphi$ as $\neg\Box\neg\varphi$.

Models $M = \langle W, R, V \rangle$ consist of a set of worlds W, an accessibility relation R, and a valuation function V assigning to each propositional variable P and world w a value $V(P, w) \in \{t, f, b, n\}$.

2.1.2 SEMANTIC CLAUSES

The truth conditions for formulas at a world w in model M are defined recursively:

1. $M, w \models v(P) = x$ iff $V(P, w) = x$, where P is a propositional variable.
2. $M, w \models \Box\varphi$ iff for all w' such that wRw', $M, w' \models \varphi$.
3. $M, w \models \neg\varphi$ iff $M, w \not\models \varphi$
4. $M, w \models \varphi \wedge \psi$ iff $M, w \models \varphi$ and $M, w \models \psi$
5. $M, w \models \varphi \rightarrow \psi$ iff $M, w \not\models \varphi$ or $M, w \models \psi$

For formulas φ other than valuation atoms:

6. $M, w \models v(\varphi) = t$ iff $M, w \models \varphi$
7. $M, w \models v(\varphi) = f$ iff $M, w \not\models \varphi$
8. $M, w \models v(\varphi) = b$ holds at no world.
9. $M, w \models v(\varphi) = n$ holds at no world.

Clauses 8 and 9 follow from 6 and 7: since the satisfaction relation is bivalent (for any φ and w, either $M, w \models \varphi$ or $M, w \not\models \varphi$), exactly one of $v(\varphi) = t$ and $v(\varphi) = f$ holds at each world. The values b and n are therefore inapplicable to formulas other than valuation atoms. This asymmetry of being four-valued for atomic propositions and bivalent for complex formulas does not affect the framework's argumentative work: modus tollens at the modal level operates on the bivalent satisfaction of $\neg\Box$-formulas,

regardless of whether the underlying atomic variable carries a four-valued assignment. The four-valued apparatus permits paraconsistent valuation of object propositions but does not enter the inferential chain that refutes necessity.

2.1.3 DEFINITION

The modal catuṣkoṭi formula T(P) is:

$$T(P) =_{def} \neg\Box(v(P) = t) \wedge \neg\Box(v(P) = f) \wedge \neg\Box(v(P) = b) \wedge \neg\Box(v(P) = n)$$

T(P) asserts: no truth value is necessarily assigned to P.

2.2 SELF-APPLICATION

Importantly, the formula applies to itself. As T(P) is itself a proposition, T can be applied to T(P), like so:

$$T(T(P)) = \neg\Box(v(T(P)) = t) \wedge \neg\Box(v(T(P)) = f) \wedge \neg\Box(v(T(P)) = b) \wedge \neg\Box(v(T(P)) = n)$$

This asserts: the truth value of the claim "P has no necessary truth value" is itself not necessarily any of the four values.

This process can continue reflexively as T(T(T(P))), and so on, without a halting point. Self-application is well-formed by formation rule 2, and the truth conditions for v(T(P)) = x are given by semantic clauses 6–9. Because T(P) asserts something about V(P) - not about V(T(P)) - no liar-like circularity arises. The non-terminating regress reflects the structure of the analysis, not paradox.

This prevents the formula from stabilizing into a necessary meta-position. One cannot assert 'all propositions lack necessary truth values' as a necessary truth, as that assertion is itself contingent - true in some models, false in others. This establishes the non-necessity as structural rather than a defect to be remedied.

More precisely: for any natural number n, the n-fold self-application $T^n(P)$ is satisfiable in modal systems that admit asymmetric accessibility: K, T, S4, and their non-normal extensions. At each stage, the formula is contingent in these systems, and the next application therefore has the semantic space to operate.

Crucially, n-fold self-application fails to be satisfiable in S5. This restriction is principled rather than arbitrary: S5's equivalence relation forces modal claims to be class-constant within equivalence classes, which collapses the directional structure that grounding relations require. That is, if worlds can access each other's alternatives, the truth of necessity-claims becomes fixed across them uniformly. Since *svabhāva* is precisely a grounding-style necessity – necessity in virtue of intrinsic nature, independent of conditions – it is not appropriately modeled by the uniform necessity of S5. The formula's failure to iterate in S5, rather than being a defect, is a formal registration of the incompatibility between uniform necessity and the asymmetric dependence relations characteristic of *svabhāva*. The n-fold satisfiability result for the

appropriate modal systems is formally verified in Lean 4 (see Appendix). This accommodates Nāgārjuna's rejection of all four lemmas so far as they relate to metaphysical truth values, as rejection compounds indefinitely.

This self-referential structure requires that T(P) remain within the scope of the analysis rather than quarantined at a separate metalinguistic level. Such refusal of type-theoretic stratification is standard in paraconsistent treatments of self-reference (Priest 1987) and is philosophically necessitated: a rigid level-hierarchy would presuppose the stable ground the formula shows to be unavailable.

Formulas of the modal language receive values in {t, f} only, as a consequence of semantic clauses 6–9; the four values {t, f, b, n} apply to propositional variables through the base valuation V. When T applies to modal formulas like T(P), the conjuncts concerning b and n are trivially satisfied, and T(T(P)) reduces to classical contingency: T(P) is neither necessarily true nor necessarily false. This asymmetry is philosophically appropriate: the method operates within conventional discourse, which is bivalent; what the method reveals is that its object lacks necessary truth value. This mirrors the Two Truths structure already invoked: conventional reasoning about ultimate emptiness.

2.3 RELATION TO PRIOR FORMALIZATIONS

Graham Priest in *In Contradiction* (1987) articulated and defended the principle of dialetheism, in which the negation of a true statement can also be true in non-classical logical systems, such as paraconsistent logics. Priest has since elaborated extensively on paraconsistent logics, notably in *An Introduction to Non-Classical Logic* (2001).

In "The Logic of the Catuskoti" (2010), Priest applied this framework directly to Nāgārjuna's *tetralemma*, demonstrating how paraconsistent four-valued semantics can accommodate the apparent contradictions within the *catuṣkoṭi* without triggering the principle of explosion.

However, Priest's formalization faces a difficulty. Nāgārjuna's argumentative method in the *Mūlamadhyamakakārikā (Root Verses of the Middle Way)* relies extensively on *modus tollens* to frame several series of *reductio ad absurdum* arguments - demonstrating that if X had *svabhāva*, absurd consequences would follow, and therefore X lacks *svabhāva*. Under Priest's FDE (First Degree Entailment), *modus tollens* is invalid: from $P \rightarrow Q$ and $\neg Q$, one cannot infer $\neg P$ when P may have the value 'both.'[1]

The modal extension offered here resolves this tension. By situating the four values within a modal framework, the reasoning *about* necessity claims operates classically.

[1] Priest (1987, Ch. 6) develops a strict entailment connective → that preserves modus tollens. Priest explicitly declines to import this connective into his 2010 *catuṣkoṭi* formalization as "irrelevant for the matter at hand" (Priest 2010, 34 n. 1). Doing so would require a designated world G with the omniscience condition ∀w ∈ W, wRG, non-reflexivity of R elsewhere, or in the updated account (Priest 2006, §19.8) impossible worlds with a ternary relation and an Augmentation Constraint Priest himself calls "not, in general, effectively checkable" (Priest 2006, 272). Our modal approach avoids all of these in standard Kripke frames.

Modus tollens is preserved at the level where Nāgārjuna actually employs it.[2] Of course, we do not claim Nāgārjuna reasoned in terms of possible worlds; we are claiming his arguments exhibit a structure that modal logic makes perspicuous, just as formalizing any natural-language argument in logic doesn't impute the formalism to the original author.

It should be noted that the formula does not impose modal structure on non-modal arguments. *Svabhāva* is constitutively modal: to possess *svabhāva* is to possess one's nature independently, intrinsically, not in dependence on conditions - which is to possess it necessarily. When Nāgārjuna concludes that X lacks *svabhāva*, he concludes that X does not have its nature necessarily. The formula makes explicit the modal structure already present in *svabhāva*-discourse rather than adding something foreign to the original arguments.

This interpretation is not merely one option among equals. Nāgārjuna demonstrably employs *modus tollens* throughout the MMK. Readings that treat *svabhāva* non-modally render these arguments invalid under four-valued semantics. The modal reading preserves them. The standard of charitable interpretation in which we attribute logical competence to the author favors the reading under which Nāgārjuna's explicit argumentative moves remain valid.

For instance, consider MMK 2:5-6. Nāgārjuna argues: if the act of going inherently exists in the path being traversed, then two acts of going would follow - the going that makes the path 'being traversed,' and the going supposedly located there (Siderits & Katsura 2013, 34-35) (Translating from the original Sanskrit). Two acts of going entail two goers. But two goers are absurd; there is one motion and one mover. Therefore, going does not inherently exist in the path.

Reconstructed modally, representing entailments that preserve necessity:

(1) $\Box(v(\text{going-in-path}) = t) \rightarrow \Box(v(\text{two-goings}) = t)$
(2) $\Box(v(\text{two-goings}) = t) \rightarrow \Box(v(\text{two-goers}) = t)$
(3) $\neg\Box(v(\text{two-goers}) = t)$ [there is contingently one goer; necessity fails]
(4) $\therefore \neg\Box(v(\text{going-in-path}) = t)$

Modus tollens applies twice, classically, at the modal level. The four-valued structure permits the object-level propositions to receive any value; what the argument refutes is the necessity of those assignments. This is the argumentative pattern throughout the MMK.

3. SYSTEM-AGNOSTICISM

We demonstrate that the formula is satisfiable across the standard hierarchy of normal modal logics.

[2] This also reflects the “Two Truths” in Madhyamaka Buddhist philosophical thought, in which one may speak of the “conventional” (classical reasoning) to frame modal discussion of the “ultimate” nature of existence. Applied reflexively and given no halting condition, the two frames interleave without prioritization; the analysis becomes the analysandum.

3.1 SATISFIABILITY IN K

Consider the following Kripke model $M = \langle W, R, V \rangle$:

- $W = \{w_0, w_1, w_2, w_3, w_4\}$
- $R = \{(w_0, w_1), (w_0, w_2), (w_0, w_3), (w_0, w_4)\}$
- $V(P, w_1) = t$; $V(P, w_2) = f$; $V(P, w_3) = b$; $V(P, w_4) = n$

At w_0, for each value $x \in \{t, f, b, n\}$, there exists an accessible world w_i such that $V(P, w_i) \neq x$. $V(P, w_0)$ may be assigned any value; the accessibility to w_1–w_4 provides the necessary counterexamples regardless. Therefore:

- $\neg\Box(v(P) = t)$ holds at w_0 (since $V(P, w_2) \neq t$)
- $\neg\Box(v(P) = f)$ holds at w_0 (since $V(P, w_1) \neq f$)
- $\neg\Box(v(P) = b)$ holds at w_0 (since $V(P, w_1) \neq b$)
- $\neg\Box(v(P) = n)$ holds at w_0 (since $V(P, w_1) \neq n$)

As the conjunction holds, the formula is satisfiable in K.

3.2 EXTENSION TO T, S4, S5

From the preceding, one then sees how the formula remains satisfiable in T, S4, and S5 modal logics.

- Under T in which $\Box\varphi \rightarrow \varphi$, the additional reflexivity adds accessible worlds and does not remove them, keeping the formula satisfiable.
- Under S4 in which $\Box\varphi \rightarrow \Box\Box\varphi$, the addition of transitivity chaining world accessibility does not affect the formula, as transitivity governs which worlds access which and does not touch the values those worlds contain. In other words, the transitivity of accessibility does not force the fixity of any value.
- Under S5, the strongest system, in which $\Box\varphi \leftrightarrow \Diamond\Box\varphi$ and $\Diamond\varphi \leftrightarrow \Box\Diamond\varphi$, worlds partition into equivalence classes with full mutual accessibility. However, mutual accessibility does not entail uniform valuation. A model with diverse valuations across worlds within a single equivalence class remains well-formed and satisfies the formula. The model constructed for K above, with R extended to an equivalence relation over W, satisfies the formula in S5. While T(P) itself is satisfiable in S5, the iteration T(T(P)) is not; this restriction and its philosophical significance are discussed in Section 2.2. (See Appendix for formal verification of T(P) satisfiability in K, T, S4, S5, and satisfiability of T(T(P)) in K, T, S4.)

3.3 NEIGHBORHOOD AND NON-NORMAL SEMANTICS

Non-normal semantics are weaker than the above systems in that they validate fewer necessity inferences. Intuitively, if the formula is satisfiable in the minimal normal modal logic of K, it remains satisfiable in systems more permissive toward when necessity fails. For example, in neighborhood semantics, $\Box\varphi$ holds at w iff the

proposition expressed by φ belongs to the neighborhood N(w) – a designated set of propositions deemed necessary at that world. If models exist in K where no value is necessitated, analogous models exist in neighborhood semantics where no value-assignment proposition belongs to the neighborhood of necessities.

Similarly, non-normal modal logics weaken the inference rules governing □. Since our result requires only that necessity claims *can* fail, not that they *must*, weaker systems preserve satisfiability.

A full treatment is beyond the scope of this paper, but we assert the structural element: the formula's satisfiability does not depend on features peculiar to normal modal logic.

3.4 SIGNIFICANCE

By maintaining satisfiability across a broad array of modal systems, both normal and non-normal, the formula does not merely hold as an artifact of system choice. This is due to the nature of modal systems: refutation of necessity makes minimal commitments, and systems generally distinguish themselves by which positive necessity claims they validate, not by which necessity-denials they permit.

The iteration's sensitivity to S5, discussed in Section 2.2, does not undercut this. On the contrary: the iteration holds in every modal system that admits asymmetric accessibility and fails only in the system that imposes the uniform necessity *svabhāva* is defined against. The instability the formula identifies is a structural feature across exactly the formal choices appropriate to modeling dependence. This parallels the pervasiveness of *śūnyatā* on all levels of conceptual analysis where dependence operates. A Lean 4 formalization verifying these results is detailed in the Appendix.

4. ADDRESSING OBJECTIONS

A position should be considered and defended from the viewpoint of the strongest identifiable objections. These are identified as follows:

4.1 THE TRIVIALISM CONCERN

The formula and its implications might be read at first glance as overly nihilistic in suggesting non-necessity to all truth value claims. However, it must be noted that the formula does not affirmatively deny the presence of valuations and is solely focused on the modal status of such valuations. This parallels the Buddhist concession that conventional function is not nonexistent. The nihilist watchword of "anything goes" is subjected to the formula as a matter of course and revealed to be not necessary, allowing for pragmatic response.

4.2 THE TRANSCENDENTAL OBJECTION

It is possible for a critic to attempt to attack the formula on what seems to be its own

grounds by noting that to state the formula presupposes semantic stability. However, we assert that it does not presuppose stability in the sense of refusing or failing to subject that stability to analysis and refutation. Instead, the observation is embraced as an unavoidable consequence of the structure: the formula, applied to itself, is not necessarily true in all possible worlds. Indeed, this unavoidable presupposition of stable referents, fixed meanings, and determinate boundaries is precisely what Madhyamaka identifies as *svabhāva*. Every proposition, insofar as it functions semantically, encodes this presupposition. The formula therefore applies to any proposition, not merely to explicitly metaphysical claims.

This relates to the 'no thesis' statement of Nāgārjuna, to be discussed in section 5: the critic's own assertion that semantic stability is necessarily presupposed is itself a necessity claim, subject to the same analysis.

4.3 REFLEXIVE SCOPE

A unique element of the formula's operation is that any objection to the formula must itself be expressible as a proposition; any such proposition is subject to the formula's analysis. Critics engaging with this framework are therefore obligated to specify what kind of claim their objection is making and why that claim is not reachable by the operation it purports to refute. Objections that cannot meet this requirement have not engaged the framework, as they have instead asserted its inadequacy without showing the inadequacy's content. Indeed, objections, in their structure, create binary divides between permissibility and impermissibility, or accessibility and inaccessibility. Such propositional binaries are how *svabhāva*-like necessity claims are formed in conventional speech. As such, dialectical engagement with necessity-enforcing binaries, such as through Nāgārjuna's use of *reductio ad absurdum*, collapses them into incoherence.

5. THE VIGRAHAVYĀVARTANĪ DEFENSE

Nāgārjuna defends his approach, as outlined in the *Mūlamadhyamakakārikā*, in the *Vigrahavyāvartanī*. A much-discussed passage within the *Vigrahavyāvartanī* is as follows:

> 29. If I had any thesis, that fault would apply to me. But I do not have any thesis, so there is indeed no fault for me.
> If I had any thesis, the earlier fault you mentioned would apply to me, because the mark of my thesis has been affected. But I do not have any thesis. To that extent, while all things are empty, completely pacified, and by nature free from substance, from where could a thesis come? In this context your statement "there is precisely that fault for you, because the mark of your thesis has been affected" is not tenable (Westerhoff 2010, 30).

This passage has long vexed commentators on Nāgārjuna because it seems to itself assert a thesis – the thesis of not having a thesis. Scholars have proposed semantic

(Westerhoff 2010; Siderits & Garfield 2013), pragmatic (Tillemans 2016), and dialetheist (Deguchi, Garfield & Priest 2008) readings, among others.

However, each proposed reading has generated its own tensions and logical issues in turn. As previously discussed, dialetheist and paraconsistent readings cannot easily accommodate Nāgārjuna's extensive usage of *modus tollens* in argument. Semantic readings distinguish levels of discourse between the conventional and ultimate truth without claiming Nāgārjuna rejects assertions, but in so doing risk either infinite regress in search of ground ("The ultimate truth is that there is no ultimate truth"), or make 'conventional' speech dependent on a metaphysically existent "real" to function (Ferraro 2013). [3] Pragmatic readings emphasize Nāgārjuna's usage of *reductio* arguments (known in Indian philosophy as *prasaṅga*) to refute positions without offering theses, but lack the formalization to offer justification of the status of his "no thesis" statement.

Our formula neither asserts a metaphysical existent nor evades the content of Nāgārjuna's maneuver. A subtle distinction must be highlighted: Semantic readings must take infinite regress as a problem due to their reliance on positive assertions regarding conventional and ultimate truth, which presupposes a stable ground from which no more assertions can productively be made, indicative of *svabhāva*. In this sense, the circularity is vicious.

Our formula is non-vicious, or virtuous; ground is not sought, because the formula's regress demonstrates ground is unavailable.

More directly: A regress is vicious when each stage requires completion of subsequent stages for its justification; absent termination, no stage is justified. The formula's regress is virtuous because no stage claims justification from subsequent stages. Each application of T is self-contained. The infinite continuation demonstrates a structural feature (the unavailability of ground) rather than constituting a failure to find it.

In this sense non-termination is the *result* rather than a defect to be remedied. Non-termination prevents stabilization into a meta-position and parallels the Buddhist doctrine of the conceptual proliferation of dualisms within the mind. Nāgārjuna does not seek to step outside of this proliferation in order to assert a metaphysical existent that lies outside of mind. Rather, lacking the specific formalism seen in this paper, he applies the *method* of the formula to all *svabhāva*-propositions so as to deconstruct the processes of conceptual thought within the inescapable language of conceptual thought.

6. IMPLICATIONS

Our formula directly translates Nāgārjuna into the language of formal modal logic within a Western analytical lens. It uses the tools of modal and classical logic in order

[3] Siderits & Garfield (2013) reply to Ferraro by parameterizing 'ultimate truth,' but it should be noted this disambiguation itself constitutes a semantic distinction subject to modal analysis; "The very idea of how things ultimately are is incoherent" is an assertion with a truth value (Siderits & Garfield 2013, 5). The formula provides what parameterization cannot: structural non-stabilization without necessarily requiring either endless disambiguation or embracing true contradictions.

to reject the foundationalist metaphysics that underlie classical reasoning in Western philosophical traditions. This is itself *prasaṅga* in action – the formula uses the master's own tools to reveal the instability of the master's house. Our approach highlights that the "East vs. West" division in philosophical circles is an arbitrary convention given the existence of this formula, which can be parsed in a system-agnostic fashion and reflected equivalently in human language.

Work remains to be done to investigate whether this formula holds in application beyond Nāgārjuna himself. However, it brings significant clarity to some of the most disputed passages within the *Mūlamadhyamakakārikā* (MMK). Three are hereby given:

"Emptiness is taught by the conquerors as the expedient to get rid of all [metaphysical] views. / But those for whom emptiness is a [metaphysical] view have been called incurable." (MMK 13:8; Siderits & Katsura 2013, 145)

Here, while the context is more pedagogically oriented, the concern is similar to that of the 'no thesis' proposition and is resolved in the same fashion by the formula: Nāgārjuna is not making a metaphysical assertion but is instead expressing one step of the self-reflexive *prasaṅga* methodology. His defense in the *Vigrahavyāvartanī* expounds on this.

"All is real, or all is unreal, all is both real and unreal, all is neither unreal nor real; this is the graded teaching of the Buddha." MMK 18:8 (ibid, 200)

On paper, this seems in contradiction to the formula, as Nāgārjuna seems to be expressing a positive assertion of all four lemmas being existents. However, operative here is the descriptor that "this is the graded teaching of the Buddha" – The Buddha employs pedagogical structure as skillful means that does not contradict the formula, because a teaching *does not necessarily impart* a metaphysical existent, nor is it *itself necessarily true*. More directly, all four lemmas are held to be *available* rather than *necessary*.

> Moreover, the objection that you make concerning emptiness cannot be a faulty consequence for us or for emptiness.
> All is possible when emptiness is possible.
> Nothing is possible when emptiness is impossible.
> You, throwing your own faults on us,
> are like the person mounted on a horse who forgets the horse.
> If you look upon existents as real intrinsically,
> in that case you regard existents as being without cause and conditions.
> Effect and cause, as well as agent, instrument and act,
> arising and ceasing, and fruit – all these you thereby deny.
> Dependent origination we declare to be emptiness.
> It [emptiness] is a dependent concept; just that is the middle path.
> (MMK 24:13-18) (*ibid*, 276-277)

Here, Nāgārjuna denies an intrinsic necessity to emptiness, dependent origination, and the Middle Way itself. As such, the Way rejects a position between extremes or further fifth truth-value outside of the *tetralemma*, as Priest has suggested to resolve

the paradox of ineffability (*The Fifth Corner of Four*, 2018).[4] The inexhaustible nature of the formula expresses the dependency that characterizes dependent origination.

We can see from these examples that the formula's framework of modal non-necessity coherently ties together multiple difficult passages of the MMK into a unified interpretation. Whether this coherence extends to subsequent transmissions, such as into *Chan* Buddhism and Tibetan scholastic traditions, remains to be investigated.

7. CONCLUSION

This paper has presented a modal extension of Belnap-Dunn semantics that formalizes the Madhyamaka catuṣkoṭi while preserving *modus tollens*. The formula's self-application structurally instantiates Nāgārjuna's 'no thesis' defense without evasion or paradox within the modal systems appropriate to asymmetric dependence. The base formula's satisfiability across K, T, S4, and S5 demonstrates that the core result is not an artifact of formal choices, while the iteration's restriction to non-uniform modalities reveals that S5's uniform necessity is structurally incompatible with the directional grounding characterizing *svabhāva* – the very target of Nāgārjuna's analysis.

Madhyamaka, rendered in the language of Western logic, awaits further investigation when extended to *all* propositions, given that semanticity depends on the use of stable referents, fixed meanings, and determinate boundaries that must presuppose *svabhāva* in order to perform productive work.

APPENDIX: LEAN 4 VERIFICATION SUMMARY

A Lean 4 formalization of the results below, generated and verified using Aristotle (Harmonic), is publicly available at https://github.com/rollofthedice/modal-catuskoti-lean. Aristotle is an AI theorem-proving tool that generates formal proofs that are verified in Lean 4, a proof assistant whose kernel mechanically checks each logical step. All proofs are machine-checked with no unproven assumptions.

1. The formation rules and semantic clauses of Section 2.1 are formally encoded, including the two-tier evaluation structure in which formulas other than valuation atoms receive only classical truth values.
2. T(P) is satisfiable in K, T, S4, and S5. A single two-world model with universal accessibility – which is itself an S5 frame – suffices to establish satisfiability across the entire hierarchy.
3. T(T(P)) is satisfiable in K, T, and S4. The proof uses a combined-model construction that joins two component models at a fresh root world, where one component satisfies T(P) and the other refutes it. The construction preserves

[4] While Priest (2018) resolves the ineffability paradox via a designated fifth semantic value (ineffability) in a plurivalent lattice, we argue that the modal approach remains closer to the Madhyamaka spirit of *prapañca* (conceptual proliferation). Rather than positing a static 'ineffable' value, the modal formula models the dynamic failure of analysis to settle, treating the lack of a halting point as the logical feature corresponding to emptiness. In so doing, ineffability avoids reification.

reflexivity and transitivity as needed, but does not impose symmetry.

4. For all natural numbers n, the n-fold self-application $T^n(P)$ is satisfiable in K, T, and S4. The proof proceeds by induction: at each stage, the combined-model construction from (3) is applied to a satisfying model and a refuting model of the previous level, yielding a model for the next level that preserves the relevant frame conditions.
5. T(T(P)) is unsatisfiable in S5. The proof proceeds by showing that T(P), as a conjunction of ¬□-formulas, is class-constant under S5's equivalence relation on worlds; therefore, no single equivalence class can contain both a world where T(P) holds and a world where it fails, which contradicts the requirements of T(T(P)). This result is discussed philosophically in Section 2.2.
6. The b and n conjuncts of T(T(P)) hold trivially at every world with at least one accessible world, because the valuation of a formula other than a valuation atom never yields b or n.

ACKNOWLEDGEMENTS

I thank the anonymous reviewers of the journal *Comparative Philosophy* for their constructive and insightful feedback.

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